Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

RESEARCH FUNDING AGREEMENT
Between and among
THE COMMISSION ON HIGHER EDUCATION, THE UNIVERSITY OF THE PHILIPPINES,
THE MINDANAO STATE UNIVERSITY - NAAWAN, THE PARTIDO STATE UNIVERSITY,
THE SORSOGON STATE UNIVERSITY, and THE UNIVERSITY OF EASTERN PHILIPPINES

COMMISSION ON HIGHER EDUCATION
PHILIPPINE-CALIFORNIA ADVANCED RESEARCH INSTITUTES (PCARI) PROGRAM
"Leading the Advancement of Knowledge in Agriculture and Sciences (LAKAS) Program"
("LAKAS Program")

Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

This project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal. The major output of this project is a handbook of seaweeds outlining their phytochemical properties, which will be made available to the Bureau of Fisheries and Aquatic Resources (BFAR), seaweed farmers, and other stakeholders involved in the seaweed industry as a guide for mariculture activities. The publication of a handbook of seaweeds outlining its phytochemical properties will have societal impacts in long-, medium-, and short-term scales. For long-term impact, the knowledge gained from this project will serve as the starting point of future efforts into product development for various applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals. For medium-term impact, identification of seaweed species that are more profitable for a specific location will guide seaweed farmers on what species they will prioritize for farming and improvement of mariculture methods, which may result in poverty alleviation in coastal communities. For short-term impact, the identification of seaweed species with high nutritional content and safe for direct consumption will provide additional or alternative food sources for the general population, who are clamoring for nutritious dietary sources. The project is envisioned to contribute to knowledge generation about potential uses of seaweeds that will have societal and economic impacts to coastal communities and the Philippine population in general.

The specific objectives of the proposed project are:
1. To perform systematic phytochemical characterization (including lipid, fatty acid, amino acid, and total phenolic content; proximal analysis; β-carotene, fucoxanthin, fucosterol, and iodine content of at least 50 seaweed species from the Philippines;
2. To assess the potential anti-cancer activity of extracts and purified compounds from seaweeds and associated epiphytes; and
3. To conduct training workshops and seminar on seaweed farming techniques, seaweed collection and preservation methods, and seaweed propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders.

The partner HEIs will actively participate in the conduct of research activities from site selection, sampling, sample processing and analysis, data collection and analysis, manuscript and report writing, and in training/workshops. The specific roles and participation of partner HEIs are outlined in the document "IHITM_2018 035_Summary of Role and Participation of HEIs_PhycoPRO 26 May 2022."

Total Budget Requirement: PhP 183,054,284.54
US: UCSD: PhP 27,464,127.74

PH:
UP-MSI: PhP 110,223,004.80
UEP, Samar: PhP 8,984,288.00
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Project Title:

Phytochemical characterization of macroalgae for food and high value products (PhycoPRO)

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Project Duration:

Two years

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I. Project Approach

A. Abstract

Seaweeds comprise one of the most important biotic components of different marine ecosystems, including estuaries, mangrove swamps, salt marshes, and coral reefs. They also serve as the base of the marine trophic food chain and are often used as an indicator of ecosystem health. The geographical distribution of seaweeds is largely influenced by physical, chemical, and biological factors. Among biological factors, the presence of epiphytes constitutes a major role in the growth and development of seaweeds because epiphytes secrete metabolites that are critical for the survival of the host but may be harmful in different circumstances. Centuries of human consumption have resulted in diverse uses of seaweeds from different algal groups including red (Rhodophyta), brown (Phaeophyceae), and green (Chlorophyta) algae, which represents a rich resource for a multitude of applications including products for food, nutraceuticals, pharmaceuticals, and other bioactive compounds. The worldwide seaweed industry relies on the seaweed produced by Chile (39%), China (23%), Norway (13%), Japan (9%), France (4%), Ireland (2.73%), Iceland (2%), Canada (1%), and South Africa (1%) (West et al. 2016).

Seaweeds (=macroalgae), as processed and unprocessed food, have an annual commercial value of several billion dollars. Seaweed extracts are commonly used in preparing foods and the direct consumption of seaweed has existed for centuries in the diets of East Asian and Pacific Island societies (Arasaki & Arasaki 1983). The red seaweed nori, or laver (i.e. ***Porphyra*** spp. and ***Pyropia*** spp.), is one of the most important commercial food seaweeds. In Japan, approximately 100,000 hectares of shallow bays and seas are farmed for nori production. Several species of ***Pyropia*** are found in Northern Philippines (Ilocos Norte, Cagayan, Isabela, and Batanes) where wild stocks are harvested for local consumption (Dumilag & Montilla 2018; Dumilag et al. 2017).

Species of ***Sargassum*** (a type of brown seaweed) are also harvested from wild beds along rocky shores, particularly in Ilocos Norte and Davao, where they may be eaten with fish and in soups. The green seaweed ***Ulva*** look somewhat like leaves of lettuce (sea lettuce) and are eaten as salads or in soups, relishes, and meat or fish dishes. Other seaweed species harvested and prepared as salads are the brown seaweed ***Hydroclathrus***, the green seaweed ***Caulerpa*** (sea grapes, common name "*lato*"), and several red seaweeds such as ***Gracilaria***, ***Kappaphycus***, and ***Eucheuma*** (Montaño et al. 2015).

The cell walls of many seaweeds contain phycocolloids that can be extracted by hot water. The three major phycocolloids are alginates, agars, and carrageenans (Stiger-Pouvreau et al. 2016). Alginates are extracted primarily from brown seaweeds, and agar and carrageenan are extracted from red seaweeds. These phycocolloids are polymers of chemically modified sugar molecules, such as galactose in agars and carrageenans, or organic acids, such as mannuronic acid and glucuronic acid in alginates. Most phycocolloids can be safely consumed by humans and other animals, and many are used in a wide variety of prepared foods, such as ready-mix cakes, instant puddings and pie fillings, and artificial dairy toppings (Stiger-Pouvreau et al. 2016). Alginates, or alginic acids, commercially extracted from brown seaweeds are used in ice creams to limit ice crystal formation (producing a smooth texture), in syrups as emulsifiers and thickeners, and in candy bars and salad dressings as fillers (Stiger-Pouvreau et al. 2016). Alginates can be extracted from Philippine ***Sargassum***, ***Hormophysa*** and ***Turbinaria***.

Agars extracted primarily from species of red seaweeds, such as Philippine ***Gelidium***, ***Gracilaria***, ***Gelidiella***, and ***Pterocladia***, are used in instant pie fillings, canned meats or fish, and bakery icings and for clarifying beer and wine. Agar is also used extensively in laboratory research as a substrate for growing bacteria, fungi, and algae in pure cultures and as a substrate for eukaryotic cell culture and tissue culture (Stiger-Pouvreau et al. 2016). Carrageenans are extracted from various red algae, including ***Eucheuma*** and ***Kappaphycus*** in the Philippines. Carrageenans are used for thickening and stabilizing dairy products, imitation creams, puddings, syrups, and canned pet foods and are used in the manufacture of shampoos, cosmetics, and medicines (Stiger-Pouvreau et al. 2016).

For many centuries, seaweeds around the world have been widely used as agricultural fertilizers. Coastal farmers collect seaweeds by cutting them from seaweed beds growing in the ocean or by gathering them from masses washed up on shores after storms. The seaweeds are then spread over the soil. Dried seaweed, although almost 50 percent mineral matter, contains a large amount of nitrogenous organic matter. Commercial extracts of seaweed sold as plant fertilizers contain a mixture of macronutrients, micronutrients, and trace elements that promote robust plant growth (Delaney et al. 2016). Seaweeds synthesize primary and secondary metabolites. These bioactive compounds can play a major therapeutic role in disease prevention in humans (Déléris et al. 2016). These compounds include polysaccharides, pigments, fatty acids, polyphenols and peptides that have been proven to possess various beneficial biological properties that could potentially contribute to functional food and nutraceutical development. The functional traits of these bioactive compounds include

antioxidant, antibacterial, anticancer, antidiabetic, antitumor, antiviral, anti-inflammatory and anticoagulant properties (Vonthron-Sénécheau 2016).

The cosmetics industry is a key industrial sector, which is worth more than €425 billion worldwide. This industry is always searching for new ingredients and had been looking from the richness and diversity of the marine world. Seaweeds are rich in bioactive compounds and are exploited as functional ingredients for cosmetic applications (Couteau & Coiffard 2016). Marine cosmetics, also known as phycocosmetics, are an economic reality and specific brands such as Algotherm, Phytomer, Daniel Jouvance, Science & Mer, or Thalgo focus on marine resources.

Seaweeds are of interest to the cosmetics industry not only as a source of minerals, polysaccharides, proteins, and lipids, but also because of the secondary metabolites such as phenolic compounds, terpenoids, halogenated compounds, sulfur derivatives, and nitrogen derivatives that they can produce. The cosmetic potential of different bioactive compounds found in seaweeds includes seaweed extracts as active ingredients for (1) slimming products, (2) moisturizing and antiaging products, (3) topical UV photoprotection, (4) skin whitening, and source of (5) excipients, (6) preservatives, (7) dye and pigment, and (8) aromas and fragrances (Couteau & Coiffard 2016).

The Philippines has an economic advantage in the global seaweed industry with an available farmable area of ~200,000 hectares along the coastlines. Currently, roughly 12% of this area is utilized for farming of seaweed species and strains from 4 genera namely: *Kappaphycus*, *Eucheuma*, *Gracilaria* and *Caulerpa.* The Philippines is host to 1079 identified seaweed species and may still have many unidentified species to be catalogued. Seaweed farming sustains the livelihood of more than 200,000 fishing families and generates about 250 million US dollars in export earnings (SIAP, 2017). The export destination includes Australia, China, South Korea, the USA, and others, and most of the export is of carrageenan. Among the destination countries, the US accounts for about a quarter of export earnings. However, the US National Organic Standard Board (NOSB) recommended the removal of carrageenan from the National List of Additives for Organic Products. This recent development in parallel with volatility of the seaweed local price, financing constraints, and dwindling numbers of seaweed farmers, will influence the potential earnings from seaweed production (SIAP, 2017). The reliance of the seaweed industry on the importance and demand of carrageenan poses an economic problem with the current weak international demand. Given the potential economic value and increase in demand for other seaweed-derived compounds from the international market, further research into seaweeds and seaweed culturing techniques and development of processes producing high-value products are necessitated.

Aside from carrageenan, there are other important bioactive compounds derived from seaweeds including carotenoids, fatty acids, fucoidan, and phycobiliproteins, among many others. The value of the world wide carotenoid market was nearly $1.2 billion in 2010 and is projected to reach over $1.4 billion by 2018 (BBC, 2011). β-Carotene, a major carotenoid of brown algae, has the largest share of the market as a dietary supplement and in the food market. Fucoxanthin is another orange carotenoid pigment that is a product of brown algae (Muradian et al., 2015). There is a robust literature on the health benefits of diets enriched in fucoxanthin, either taken as a supplement or through consumption of the source macroalgae. Mono-unsaturated fatty acids (MUFAs) and polyunsaturated fatty acids (PUFAs) have been shown to possess a variety of health benefits and are thus identified as important components of human nutrition. The major PUFAs of algae that exhibit health benefits are docosahexaenoic acid (DHA) and eicosapentaenoic acid (EPA). Fucoidan is the name given to a heterogeneous group of sulfated polysaccharides that can be obtained from various brown algae (Atashrazm et al., 2015). The variable linkages between these sugar units, location of sulfate groups and the trace components of other sugars give these biopolymers very different biological and pharmacological properties, depending on the source organism, time of collection, and processing parameters. The reported useful biological properties of fucoidan are largely in anticancer and antiproliferation (Atashrazm et al., 2015). The distinctive pigments of red algae, known as phycobiliproteins, have a growing number of biomedical applications. These pigments, including phycoerythrin, phycocyanin, and allophycocyanin, are exceptionally bright and have been used in such applications as cell marking for use in fluorescent activated cell sorting (FACS) (Kronick, 1986). Because of their light absorbing characteristics, they have also been used in photodynamic therapies (Huang et al., 2002), as well as for their antioxidant and anti-inflammatory properties (Romay et al., 2003). Finally, there is a growing use of these pigments in the cosmetic and food industries.

The **numerous important compounds derived from seaweeds** highlight the potential of developing it as a **renewable resource for a biobased economy**. Unlike Japan, Korea, and Chile, where seaweed is commercially available as dried products for food, in the Philippines its use is mostly limited to consumption of freshly harvested seaweed, which has a very short shelf life. Moreover, there has been no systematic analysis of the food quality of Philippine seaweed species [content such as amino acids

(AA), fatty acids (FA), carbohydrates, vitamins, minerals, and heavy metals]. With these diverse seaweed species and massive untapped potential in the Philippines, **this project will focus on the phytochemical characterization of at least 50 seaweed species in the country. The phytochemical characterization of seaweed species will serve as the foundation for development of processes for seaweed utilization for food or as a source of material for nutraceuticals, which will be pursued by the proponents in future projects. The proposed widespread sampling is a great opportunity for bioprospecting to discover biomedicinal products from Philippine seaweeds through a comprehensive screening program for high value materials with a specific focus on anti-cancer agents.**

The natural products of marine organisms are of high structural diversity and are produced in biological systems for inherent biological purposes; thus, they are best screened in mechanism-agnostic screens such as the bioassays in the Gerwick laboratory at UCSD. These natural products have a high potential to effect cells in new and diverse ways of relevance to cancer chemotherapy. Certainly, diversity in discovery programs should be a high international priority given the dismal therapeutic outcomes against the major human solid tumors over the past two decades of research that have featured approaches other than those proposed here. The future of cancer treatment is envisioned to revolve around the identification of specific DNA mutations in a given cancer patient's tumor cells followed by the development of a specific cocktail of anticancer drugs "targeted' to those modifications. Unfortunately, there are few such drugs available to mix into these futuristic cocktails, and there are literally thousands of potential targets. Through projects such as the one proposed here that throw a wide net around unique natural products for their anticancer activity and then define the mechanisms of the therapeutically active ones, we have the highest probability of finding agents that work by new and specific mechanisms. These advances are bringing the future of cancer therapy into the present. It is envisioned that natural products screening capability in UP-MSI will be further developed through the training and assistance that will be performed with the involvement of the Gerwick group at UCSD.

Seaweed research in the Philippines was initiated from efforts of scientists from the UP-MSI, especially from the contributions of Drs. Gavino Trono Jr. (GCT), Edna Ganzon-Fortes (EGF), and Nemesio E. Moñtano Jr. (NEM). Dr. Trono primarily worked on marine benthic algal taxonomy, e.g., *Sargassum*, *Gracilaria*, and *Acetabularia*, among others, and on Philippine seaweed biodiversity and their utilization. Moreover, GCT worked on the biomass production and clonal propagation of different strains of commercially important eucheumatoids, the recruitment and seasonality of the standing stock of *Sargassum* species, the effects of different environmental factors on the growth and phycocolloid quality of different agarophytes, and recently on the cultivation of the red seaweed *Halymenia durvillei*. Dr. Ganzon-Fortes also worked on marine benthic algal taxonomy, e.g., *Acetabularia*, *Laurencia*, and *Hydroclathrus*, and on Philippine seaweed biodiversity. Moreover, EGF worked on different aspects, i.e., photophysiology, ecology, reproductive biology, agar yield, and quality of *Gelidiella acerosa*. EGF also worked on the physiology of diseased and healthy *Kappaphycus alvarezii*, carrageenan yield and quality of different euchuematoids, their cultivation and development of new strains from wild populations. Recently, she studied the nutrient physiology of *Padina* in Bolinao, Pangasinan. Dr. Montaño primarily worked on the chemical characterization of different phycocolloids, e.g., agar and carrageenan, and alginate extractable from different species of red and brown seaweeds, respectively. NEM also worked on different optimization methods, e.g., pre- and post-treatments to improve phycocolloid yield and quality. Recently, he is involved in screening of fucoidan content of Philippine brown seaweeds and worked on the cytotoxic activity of crude extract of *Sargassum*. **The research efforts of the pioneering scientists on seaweeds provided a sizable understanding of the current seaweed cultivars in the country. However, there is lack of knowledge about the phytochemical properties of the multitude of naturally occurring species, many of which may produce commercially viable compounds of interest. This proposed project is intended to address this knowledge gap and is specifically aimed towards identifying seaweed species that produce the highest yields of phycobiliproteins, which are of interest because of their applications in food, medical, and cosmetic applications, and wide use in research and development. Currently, the market price for C-phycocyanin is about PhP 12,000.00/10 mg and B-phycoerythrin is about PhP 15,000.00/10 mg. The high cost of these pure phycobiliproteins highlights the economic potential of seaweeds in the Philippines.**

This project is envisioned to enhance the status of the seaweed industry in the Philippines by **providing critical knowledge through a comprehensive phytochemical characterization of seaweeds to identify species that have higher economic potential and pinpoint locations where their growth will be maximal**. The major output of this project is a handbook of seaweeds outlining their phytochemical properties, which will be made available to the Bureau of Fisheries and Aquatic Resources (BFAR), seaweed farmers, and other stakeholders involved in the seaweed industry as a guide for mariculture activities. **The publication of a handbook of seaweeds outlining its**

phytochemical properties will have societal impacts in long-, medium-, and short-term scales. For long-term impact, the knowledge gained from this project will serve as the starting point of future efforts into product development for various applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals. For medium-term impact, identification of seaweed species that are more profitable for a specific location will guide seaweed farmers on what species they will prioritize for farming and improvement of mariculture methods, which may result in poverty alleviation in coastal communities. For short-term impact, the identification of seaweed species with high nutritional content and safe for direct consumption will provide additional or alternative food sources for the general population, who are clamoring for nutritious dietary sources. The project is envisioned to contribute to knowledge generation about potential uses of seaweeds that will have societal and economic impacts to coastal communities and the Philippine population in general.

Millions of marginalized small-scale fisherfolk are dependent upon seaweed farming for their livelihoods. However, they are faced with problems of declining productivity and associated incomes due to a dependence on old and outdated seaweed farming methods that limit people to very shallow (<2m) coastal waters, old varieties of seaweed that have been vegetatively propagated for the last five decades, and the impacts of climate change. These factors interact to result in low growth rates and high mortality from epiphytism and ice-ice disease. As a result, seaweed farmers are increasingly returning to fishing, resulting in unsustainable levels of exploitation that diminish natural resources and trap people in poverty. This project proposal will build upon our expertise in the production of new and high-performing *Kappaphycus* cultivars generated from spores of wild local stocks through sexual reproduction and growing them up in local seaweed nurseries for distribution to seaweed farmers. Promising new strains will be tested for carrageenan content to preselect those with the highest quality and yield. Good quality biomass will command higher buying price that will effectively increase household incomes of small-scale fisherfolk/ seaweed farmers to at least P15-20,000 per month.

This project represents the fusion of high scientific rigor and experimental innovation focused on the phytochemical characterization of various seaweeds species that will eventually lead to development of novel food and biotechnology products from Philippine marine algae and their epiphytes. As such, we expect a number of high impact international publications on the unique marine resources present in these marine algae. These publications will provide the scientific underpinnings for new utilization of some species of algae as a source of healthy foods, as well as validation of the nutritive value of several other seaweeds that are currently used as food. Potentially harmful properties of the seaweeds to human health will also be characterized, providing a holistic view to their utilization for these purposes. Complementing this, we expect to also define a number of promising lead compounds for preclinical development in biological assays focused on anticancer activities.

The specific objectives of the proposed project are:

1. To perform systematic phytochemical characterization (including lipid, fatty acid, amino acid, and total phenolic content; proximal analysis; *β*-carotene, fucoxanthin, fucosterol, and iodine content of at least 50 seaweed species from the Philippines;
2. To assess the potential anti-cancer activity of extracts and purified compounds from seaweeds and associated epiphytes; and
3. To conduct training workshops and seminar on seaweed farming techniques, seaweed collection and preservation methods, and seaweed propagation strategies for faculty and researchers of partner HEIs, farming communities, and other stakeholders.

The partner HEIs will actively participate in the conduct of research activities from site selection, sampling, sample processing and analysis, data collection and analysis, manuscript and report writing, and in training/workshops. The specific roles and participation of partner HEIs are outlined in the document "IHITM_2018 035_Summary of Role and Participation of HEIs_PhycoPRO 26 May 2022."

OUTCOME/SOCIO-ECONOMIC IMPACTS

The project is expected to contribute towards poverty alleviation initiatives by providing additional or alternative sources of income for fishing families in coastal communities. Diversified income sources from introduction of seaweed farming and trading will reduce the pressure on fisheries, which may lead to replenishment of standing stocks of critical fish species. This will have positive effects on fish catch volumes of small-scale fishermen in the long run. The potential improvement in income sources will trigger positive effects like dominos starting with better livelihood, improved health, education of children, women empowerment, and many other socio-cultural aspects that are linked to the financial burden of coastal families.

To ensure that this initiative is continued beyond the duration of the project, the partner HEIs will be guided on how to conduct the training workshops in seaweed farming to ensure that information about new strains or up-to-date farming techniques are disseminated to seaweed farmers and community organizations in their respective provinces. The continuity of training that will be carried out by partner HEIs will guarantee that issues in mariculture activities will be addressed in a prompt and adequate manner due to proximity of HEIs to the communities requiring them.

Approach

Background

Notable examples of isolated natural products from seaweeds. Human consumption of seaweeds has been a tradition for several centuries in China, Japan and Korea (McHugh, 2003). In the tropics, fresh seaweed prepared as salads is a delicacy among coastal dwellers from Indonesia, Malaysia and the Philippines. Today, the use of seaweed, a macroalgae, as food and functional ingredient is known worldwide. Macroalgae have traditionally been an extremely rich source of bioactive small molecules, some of which have potential biomedical utility. For example, others as well as ourselves have characterized small halogenated monoterpenes as possessing anticancer type properties, such as those from *Plocamium* spp. and *Portieria hornemannii*, and these have been a source of great interest by the National Cancer Institute in the US (Andrianasolo et al., 2005). Other red algal natural products have been examined for their high potential as antineoplastic agents; prominent among these is the highly active triterpene polyether thyrsiferol and related compounds from *Laurencia viridis* (Blunt et al., 1978).

The brown algae or Phaeophyceae, are similarly rich in bioactive natural products, including terpenes and meroterpenes, polyphenolics, and oxylipins. For example, we previously discovered stypoldione and related compounds, mero-diterpenoids with unique orthoquinone functionalities, to have powerful cancer cell toxicity as well as neurotoxic properties (Sabry et al., 2005). Eckol is a brown algal polyphenolic metabolite with a variety of biological activities, including inhibition of adipogenesis (Kim et al., 2017), anti-inflammation (Eom et al., 2017), and antibacterial properties (Eom et al., 2013). Finally, brown algae are rich in unique prostaglandin-like substances, known as oxylipins, and these have a variety of biological properties in mammalian systems. The egregiachlorides are one such class of unique oxylipin (Todd et al., 1993); prostanoids of this sort have some of the most potent and profound hormone-like properties.

Finally, the green algae or Chlorophyta are also a chemically rich group of marine algae that are important in tropical reef systems in a variety of ways. For example, several calcium carbonate aggregating species of the genus *Halimeda*, which are important to the construction of coral reefs, are chemically defended against predation by producing the extremely potent metabolite halimedatrial, a compound with additional biological activities such as cytotoxicity to cancer cells (Paul et al., 1988). A notable green algal metabolite of relevance to the Philippines is caulerpin which was shown to be a plant growth regulator in a root growth test of lettuce seeds and to exhibit in vitro antitumor activity (Raub et al., 1987). Caulerpin has a unique bis-indole structure which is related to the anti-inflammatory natural products topsentin (Alvarado et al., 2013) and scytonemin (Stevenson et al., 2002).

Potential of seaweed-derived high value products. Seaweeds thrive in habitats where conditions, e.g. high irradiance and ultraviolet radiation, temperature fluctuation, and desiccation among others, favor the formation of free radicals and other oxidizing agents. Their ability to minimize oxidative damage and maintain cellular integrity against these adverse conditions is related to the presence of different antioxidant systems. In living organisms, failure to activate the defense mechanism can lead to the accumulation of reactive oxygen species (ROS) and other free radicals causing irreversible damage to proteins, amino acids, lipids and DNA. In food products, ROS can promote the degradation of oil and fats contributing to the decrease in nutritional quality and security. Polyphenols and sulfated polysaccharides are among the powerful antioxidant compounds found in different seaweed species (e.g. Valentão et al., 2010; Balboa et al., 2013; Di et al., 2017). These naturally occurring compounds in seaweeds promote its use as food or as a functional ingredient that may prevent chronic diseases and improve food shelf life and quality with health-related beneficial properties (Roohinejad et al., 2017). The same compounds in seaweeds, such as cell wall polysaccharides and proteins with biomedical potentials and health benefits, also have several metal binding sites, which may contribute to the complexity of the ROS homeostasis in the cells (Kuyucak and Volesky, 1989; Reddy and Prasad, 1990; Güven et al. 1995).

Current limitations of seaweed products development. Heavy metals are naturally occurring elements of the earth's crust. Human activities including mining and smelting operations, industrial production and use, and domestic and agricultural use of metals and metal-containing compounds have drastically altered their geochemical cycles and biochemical balance (Singh et al., 2011). Heavy metal toxicity

depends on several factors such as chemical species, dose, and exposure route, as well as the age, gender, genetics, and nutritional status of exposed individuals. Arsenic (As), cadmium (Cd), lead (Pb), and mercury (Hg) are highly toxic and rank among the priority metals that are of public health significance (Tchounwou et al., 2012). Different algal polysaccharides bind with heavy metals and their binding affinity is ranked as: sodium alginate > carrageenans > agar (Güven et al. 1995). Moreover, peptides of different species of microalgae and a brown seaweed *Sargassum muticum* also bind with heavy metal through chelation (Gekeler et al., 1988). Only a few studies have conducted comprehensive investigation on different heavy metals in edible seaweeds commercialized for human consumption (e.g., Almela et al., 2006; Besada et al., 2009; Rubio et al., 2017). Considering that there is an increasing use of seaweed for food, food additives, dietary supplements, and for their medicinal properties (Liu et al., 2015; Déléris et al., 2016; Fleurence, 2016; Kang et al., 2016; Vonthron-Sénécheau, 2016), it is essential that contaminants present in different seaweed species are thoroughly investigated before they enter our daily diet.

Cyanobacterial epiphytes are especially notorious as they often go undetected on macroalgae, yet have major consequences on the chemical composition of the food material. Examples include the highly toxic polycarvernosides, products of cyanobacteria growing on red algae, and the aplysiatoxins, also products of cyanobacteria growing on red algae. In the first case, toxic outbreaks have occurred in Guam and the Philippines; in the latter location, three outbreaks of seaweed poisoning occurred from ingestion of two edible species of red alga, *Acanthophora specifera* and *Gracilaria edulis* (Yotsu-Yamashita et al., 2004). Three of 12 victims died in the first event, three of 15 died in the second event, and two of nine died in the third event, for an overall mortality of 8 deaths from 36 reported exposures (22% mortality). The causative agent was shown to be polycarvernoside A and its congeners, both in Guam and in the Philippines; however, the ultimate source of the toxins was somewhat uncertain from these studies. Subsequently, we made field collections of the free-living marine cyanobacterium *Okeania* sp. in the Caribbean, and used a cancer cell toxicity bioassay to guide the isolation of biologically active components. This resulted in the isolation of a new polycarvernoside, "D", which was closely related in structure to the previously reported polycarvernosides, and clearly showed that these metabolites are the products of marine cyanobacterial metabolism (Navarro et al., 2015).

In the Hawaiian 1994 event, consumption of the red alga *Gracilaria coronopifolia*, a seaweed which had long been safely eaten by local residents, resulted in a widespread toxic event with nausea and vomiting being the major symptom (no deaths reported). Investigation of the toxic material resulted in the isolation of a few related metabolites, clearly in the aplysiatoxin family of cyanobacterial secondary metabolites, and given the names manauealides A-C (Nagai et al., 1997). Aplysiatoxin-type metabolites are well recognized as tumor promoters due to their PKC activating properties; in the current acute poisoning situation, the toxicity appeared to be due to intestinal bleeding (Ito & Nagai, 2000). These examples demonstrate the potential for human poisonings by consumption of seaweeds that are generally safe to eat but become epiphytized by other algae and cyanobacteria, which render them toxic. **This is a principal of which the PIs of the present proposal, which is focused on exploring the nutritive value of seaweeds not yet consumed in the Philippines, are aware of. We will carefully and fully document the likelihood of seaweed species to become contaminated by epiphytes such as marine cyanobacteria, and characterize the taxonomy and chemical constituents of these contaminating species completely so as to ensure a safe food supply.**

Rationale

The diversity of seaweed species available in the Philippines and its available farmable area represent a massive potential for seaweed farming in the country. However, there are several challenges that must be addressed to maximize the potential of the seaweed industry in the country. These include:

1. Research focusing on systematic and comprehensive phytochemical characterization of seaweeds present in Philippine waters must be performed.
2. Research into potential uses and applications of seaweeds must be carried out in order to identify species that have higher economic potential and locations where their growth will be maximal.
3. Research into processes producing seaweed products must be carried out to improve their quality and stability.
4. Research focused on identification and isolation of bioactive compounds from macroalgae must be carried out to identify other potential commercial applications.
5. Research directed towards elucidating the role of associated microorganisms on the quality of seaweed-derived products and how the resident microbiome influence production of bioactive compounds and their efficacy.

This project is envisioned to address the most critical among the knowledge gaps mentioned, which is the phytochemical characterization, to enhance the status of the seaweed industry in the Philippines. This project is envisioned to enhance the status of the seaweed industry in the Philippines by providing critical knowledge through a comprehensive phytochemical characterization of seaweeds

to identify species that have higher economic potential and pinpoint locations where their growth will be maximal. The major output of this project is a handbook of seaweeds outlining their phytochemical properties, which will be made available to the Bureau of Fisheries and Aquatic Resources (BFAR), seaweed farmers, and other stakeholders involved in the seaweed industry as a guide for mariculture activities. The publication of a handbook of seaweeds outlining its phytochemical properties will have societal impacts in long-, medium-, and short-term scales. For long-term impact, the knowledge gained from this project will serve as the starting point of future efforts into product development for various applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals. For medium-term impact, identification of seaweed species that are more profitable for a specific location will guide seaweed farmers on what species they will prioritize for farming and improvement of mariculture methods, which may result in poverty alleviation in coastal communities. For short-term impact, the identification of seaweed species with high nutritional content and safe for direct consumption will provide additional or alternative food sources for the general population, who are clamoring for nutritious dietary sources. The project is envisioned to contribute in knowledge generation about potential uses of seaweeds that will have societal and economic impacts to coastal communities and the Philippine population in general.

The sampling areas that will be included in this project are the following: Catarman, Northern Samar; Dinagat, Surigao Del Norte; Tandubas, Tawi Tawi; Batan, Batanes; Bolinao, Pangasinan; Currimao, Ilocos Norte; Guiuan, Eastern Samar; Biri, Northern Samar; Iba, Zambales; Busuanga, Palawan; Sorsogon, Sorsogon; Polilio Island, Quezon; Bien Unido, Bohol; and Lian, Batangas.

The successful implementation of this project is anchored on the merging of different scientific expertise and utilization of complementary approaches in addressing the various research targets and objectives. The specific approaches and methodologies are briefly described in the following section.

Brief description of methods that will be employed in this project

1. Identification of seaweed species
   Species identification will be based on the gross morphology and identification keys provided in taxonomic handbooks for Philippine and Regional Marine Flora, e.g., Field Guide and Atlas of the Seaweed Resources of the Philippines (Trono 1997), among others. Representative samples of each species will be pressed and mounted onto an herbarium paper. They will be deposited at G.T Velasquez Phycological Herbarium, the Marine Science Institute, University of the Philippines, Diliman, Quezon City. When necessary, molecular analysis, i.e. DNA Barcoding, will be employed.

2. Optimization of standard methods for analysis of target analytes/component in seaweeds or seaweed extracts
   a. *Proximal analysis*. The water, ash, and fiber content will be determined following the standard AOAC methods (water, AOAC 934.01; ash, AOAC 930.05; fiber, AOAC 985.29).
   b. *Total lipid analysis*. The lipid content of seaweed samples will be analyzed following an optimized protocol based on the method by Folch et al. (1957) and Gosch et al. (2012). A suitable amount of dried seaweed powder will be extracted using a chloroform:methanol mixture, heated at 60°C, filtered onto pre-weighed Whatman GF/A filter, washed with NaCl solution before allowing to stand. The lower phase containing the lipids will be evaporated using a gentle stream of nitrogen prior to weighing of total lipid content.
   c. *Fucoxanthin and fucosterol analysis*. Part of the lipid extract will be used to determine fucoxanthin and fucosterol by high performance liquid chromatography following the methods proposed by Maeda et al. (2005) and the liquid extraction optimized by Fung et al. (2013).
   d. *Fatty acid extraction and analysis*. The fatty acid content of seaweed samples will be determined following a method adapted from Gosch et al (2012). Briefly, fatty acids will be methylated using a mixture of methanol and acetylchloride, heated for 1 h at 100°C, allowed to cool before addition of hexane. Complete partitioning of fatty acid methyl esters (FAMEs) will be done prior to analysis using gas chromatography mass spectrometry.
   e. *Amino acid analysis*. The amino acid content will be determined by HPLC following the method proposed by Alaiz et al (1992) and modified by Ortiz et al (2006). Briefly, hydrochloric acid will be added to a suitable amount of seaweed sample, the solution will be gassed with nitrogen, sealed, and will be incubated at 110°C for 24 h. The amino acid hydrolizate will be dried and subsequently dissolved in borate buffer prior to derivatization using diethyl ethoxymethylene malonate and determination of the amino acid derivatives using HPLC with a UV-Vis detector.
   f. *Extraction and analysis of mycosporine-like amino acids (MAAs)*. The isolation of UV-absorbing MAAs will be carried out following the protocol by Rosic et al. (2015) and subsequently analyzed using HPLC.

g. *β-carotene extraction and analysis*. The β-carotene content of seaweed samples will be extracted following the method by Norziah et al (2000) where a mixture of tetrahydrofuran:methanol followed by petroleum ether and 0.1% butylated hydroxytoluene as extraction solvents. Saponification will be carried out using a mixture of potassium hydroxide:methanol prior to drying the extract and subsequent analysis of β-carotene using reverse phase HPLC.
h. *Analysis of iodine content*. The iodine in seaweed will be derivatized with 3-pentanone and analyzed using GC-MS following the method used by Yeh et al. (2013).
i. *Determination of total phenolic content*. The total phenolic content of seaweed extracts will be determined according to the method by Wang et al. (2009) using the Folin-Ciocalteu reagent. The measurements will be done using phloroglucinol solution as calibration standard and using UV-Vis spectrophotometry.
j. *Total antioxidant activity of extracts*. The antioxidant activities will be determined based on the effect of scavenging of 1,1-diphenyl-2-picrylhydrazyl (DPPH) radicals according to the method by Souza et al (2012).
k. *Trace metal analysis*. The trace metal (essential and toxic) content of seaweed samples will be analyzed by inductively coupled plasma mass spectrometry (ICPMS). Total elemental concentration will be determined using mineralized samples while speciation analysis will be carried out on extracts using HPLC coupled with ICPMS following optimized methods modified from Rodriguez et al. (2008 and 2009).
l. *In-vitro testing to assess bioaccessibility and bioavailability of trace metals from seaweed and seaweed extracts*. An in-vitro dialyzability procedure following protocols by Moreda-Piniero et al. (2012) and Garcia-Sartal et al. (2013) will be validated for use in seaweed and seaweed extracts. These methods simulate the chemical conditions during gastric digestion.

Strategy for Meeting Milestones

To ensure that milestones will be met, the project staff that will be hired must have requisite academic training (chemistry, biology, biochemistry, molecular biology) to deliver the outputs. In addition, the equipment that will be purchased will have built-in autosamplers to maximize the number of samples that will be analyzed. Further, proper scheduling will be implemented, e.g., seaweed sampling for preservation and method optimization will be performed while procurement of equipment is being done. The details of the project activities may be found in the GANNT chart.

3. Methods for extraction, characterization, and bioactivity assays of seaweed extracts
   a. *Extraction*. The extraction of algal tissue or cyanobacterial mats or tufts will utilize our longstanding modified Bligh and Dyer method (Crews et al., 1991).
   b. *Fractionation*. Over the past 20 years, we have found it highly effective to fractionate the extract as produced above into a series of nine "pre-fractions" (e.g. fractionated before any biological assays occur). In our experience, these pre-fractions function much more effectively in biological assays because of several factors: 1) minor constituents are concentrated in the fractions such that their activity is more easily observed, 2) nuisance compounds are removed from the extract or segregated within specific fractions so they can straightforwardly be discounted, and 3) these reduced complexity materials can be directly further purified by HPLC to yield pure compounds in a single next step, thereby accelerating the bioassay-guided process.
   c. *Dereplication*. A new innovation that we have added to our natural products discovery workflow is the use of Liquid Chromatography-Mass Spectrometry with fragment ion (LC-MS2) recording on our extracts and "pre-fractions". These data are added to the massIVE database within the Global Natural Products Social (GNPS) (https://gnps.ucsd.edu/ProteoSAFe/static/gnps-splash.jsp) molecular network and then portrayed in Cytoscape. This allows known compounds to be identified early in the process, and the relationship of new compounds to known ones to be quickly realized.
   d. *Biological Assays*. In the current project, we will focus on a couple of biological assays that will reveal the presence of small molecules that i) have potential utility in the treatment of cancer, or ii) may have mutagenic or carcinogenic properties in humans if ingested as part of the diet. The first set of assays involves antiproliferative and cytotoxicity measurements using an indicator cancer cell line, the H-460 human lung carcinoma cell line from the National Cancer Institute. *Purification*. Biologically active fractions from the above evaluations will be further fractionated by HPLC to relatively pure materials, and then re-assayed in the relevant biological system.
   e. *Mechanism of Action Studies*. Ultimately, the process outlined above will lead to the biological and chemical characterization of a novel compound from various Philippine marine algae. The relationship of the new compound to known ones will be investigated through use of comprehensive databases, such as AntiMarin and SciFinder. The pharmacological mechanism

of action of the new compound can sometimes be deduced through this comparison to the literature, and for other times needs focused investigation by one of a variety of techniques with which our laboratory is experienced, for example transcriptomics, enzymatic assays and development of resistance in yeast to the isolated compound followed by sequencing.

f. *Identification of Producing Organism*. A common question that arises from the isolation of new molecules from field collected natural assemblages is who in the assembly is actually producing the desired natural product. **In the case of macrophytic marine algae, it becomes a question whether the macrophyte is the biochemical source, or is it an epiphyte, a component of the microbiome, or an endophytic symbiont such as a fungus?** To discern this is important to both public health questions as it relates to toxins in the food supply as well as to medicinal agent discovery and subsequent commercial production. In some cases this can be deduced by culture of symbiotic or associated microorganisms, as we did in the case of unusual nucleosides from the sponge *Cryptotethia crypta* (Bertin et al, 2015). Alternatively, imaging MALDI mass spectrometry is an extremely powerful method by which to demonstrate the association of a particular substance with a specific organism present in a consortium (Simmons et al, 2008). Use of genetic signatures associated with the biosynthesis of a particular molecule can also be used, as we showed in the case of the cyanobacterium *Oscillatoria spongellae* associated with the sponge *Dysidea herbaceae* (Flatt et al, 2005). Whole genome sequencing can enhance this latter approach; this is an especially effective approach when a microorganism, such as a cyanobacterial epiphyte, is suspected to be the biosynthetic source of a secondary metabolite.

Throughput of Gerwick Laboratory

Our laboratory has considerable throughput in terms of the number of samples that can be analyzed for known as well as new secondary metabolites, whether they be toxins or potential pharmaceutical yields. This comes in large part by our exceptional infrastructure with plentiful laboratory space (8 separate laboratories with space for 15-20 researchers), great instrumentation (2 LC-MS instruments with auto samplers, one GC-MS with auto sampler, and various UV, IR and polarimeters in our labs; two 500 MHz NMRs, one with an auto sampler, in our building), and well trained and experienced staff and students. With the additional resources requested under this research grant activity, we are certain that we will be able to process and evaluate the volume of materials that will derive from this collaborative project with the Philippines.

Timeline for Meeting Milestones

On the timeline for providing the information needed from our part of the project, we expect that we will at least have quarterly joint research meetings with the rest of the members of the team, and thus we will stay up to date with the processing of samples on at least a quarterly basis. Initial evaluations of materials by chemical and biological assays occur continuously, and as noted above, we have excellent facilities and personnel to work on the project. Specific deliverables from this project include technology transfer on algal extraction and fractionation, bioassay results, structures of active and inactive compounds isolated from marine algae, mechanisms of action and biosynthetic pathway deduction for a select set of bioactive compounds, and scientific outputs including publications and patents. Our research plan involves the continuous processing of materials (extraction, fractionation) to keep pace with the number of algal species examined as sources of food or other biotechnological products. Bioassays take place on a weekly basis, and instrumental analysis of samples occurs continuously as our LC-MS and NMR instruments have auto sample changers which allow samples to be run 24 hours a day, 7 days a week. With this capacity and the appropriate level of personnel, we are confident of our ability to provide deliverables as set forth in the proposal.

B. Project Team Description & Roles

The research team members comprising the PH side will be composed of two project leaders, one project staff, and four research fellows, all of whom will be involved in the planning and implementation of the project. They will be assisted by one project assistant, whose job will focus more on the administrative aspect of the work and will be devoting 100% of his time for the success of this project.

The two project leaders, Dr. Michael Roleda and Dr. Irene Rodriguez, will divide the tasks in directing the research. Dr. Roleda will spearhead all research involving seaweed collection and identification, seaweed culturing, and other related activities. Dr. Rodriguez will lead all research involving optimization and validation of standard methods for the evaluation of physical and chemical properties of seaweeds and seaweed extracts. The project staff, Dr. Lilibeth Salvador-Reyes, is also a member of the MSI faculty. Dr. Salvador-Reyes is an expert on medicinal and natural products

chemistry, biochemistry, and pharmacology. Both project leaders will be devoting 40% of their time and the project staff will be devoting 10 % of her time for the success of this project.

The science research specialists and university research associates that will be hired for this project will be identified from a pool of applicants and must have the following qualifications: a bachelor's degree in science from a reputable college or university, civil service eligibility or a license from the Professional Regulation Commission (PRC), and of good moral standing. In addition to these minimum requirements, the team members that will be chosen must demonstrate willingness to work with a team, learn new techniques and skills, work in a competitive and demanding environment, and must be interested in the pursuit of scientific excellence. All research fellows are expected to dedicate 100% of their time for the success of this project.

The Gerwick group at SIO/UCSD will be involved in teaching the Philippines group the process of organic chemical extraction and pre-fractionation of marine algae for toxin as well as medicinal agent discovery. In the Gerwick group, biological assays will be performed on these pre-fractionated materials, and then biological assay-driven isolation of active compounds will be achieved using column, flash and high-performance liquid chromatography (HPLC). Structures of bioactive compounds will be determined using spectroscopic methods including NMR and MS. Mechanisms of action of select active compounds will be explored using transcriptomics and biochemical assays. Finally, genomic data for the organisms producing the bioactive compounds will be analyzed for the presence of biosynthetic gene clusters. Deliverables from this project include technology transfer on algal extraction and fractionation, bioassay results, structures of active and inactive compounds isolated from marine algae, mechanisms of action and biosynthetic pathway deduction for a select set of bioactive compounds.

C. Annual Project Deliverables & Intermediate Key Milestones with Dates or Timeline (1-2 pages)

The specific objectives and goals of this project will be sought by the project team members simultaneously and in close coordination with each other. For the successful implementation of the project, the research team will aim to meet the following deliverables and intermediate milestones:

Deliverables	TARGET DATE
Seaweed collection for preliminary studies including optimization methods for seaweed preservation, analytical methods, bioassays, extraction methods, etc.	July 2023
Set-up of Seaweed Chem Lab	October 2023
Set of optimized methods for analysis of specific analytes (including lipid, fatty acid, amino acid, and total phenolic content; proximal analysis; β-carotene, fucoxanthin, fucosterol, and iodine content) in seaweeds and seaweed extracts	December 2023
Seaweed collection for phytochemical analysis	December 2023
Set-up of ICPMS Facility	December 2023
Set of optimized methods for analysis of specific analytes (including content of metals such as arsenic, cadmium, lead, iron, zinc, manganese, cobalt, copper, nickel, and selenium) in seaweeds and seaweed extracts	April 2023
Data and background information on the phytochemical properties of seaweed species from different sampling sites	March 2024
Seaweed collection for phytochemical analysis	April 2024
Data on compounds isolated from seaweeds with novel bioactivities	June 2024
Data on the toxicological evaluation of seaweed products	September 2024
Training/Workshop on seaweed farming	August 2024
Scientific publications, handbook	December 2024

The project research team members are also expected to present the project results in relevant international and national scientific meetings to gain feedback and insights about the work.

OUTPUTS/DELIVERABLES and CONTRIBUTION TO CHED INITIATIVES

The outputs of the PhycoPRO project will contribute to knowledge generation and economic improvement in various aspects as outlined below:

1. The **handbook of seaweeds describing their phytochemical properties and potential commercial applications** may be used in the following:

a. As a reference material for seaweed biodiversity studies by HEIs offering programs in Biology, Marine Biology, Ecology, Fisheries, and related programs;
b. As a reference material for studies on marine natural products by HEIs offering programs in Chemistry, Biochemistry, Biotechnology, and related programs;
c. As a reference material for Elementary and High School science classes to introduce students to seaweeds and its ecological or economic importance thus ensuring that the next generation of Filipinos will be educated in the biodiversity and richness of seaweed resources in the country;
d. As a reference material for education and information of people's organizations especially within coastal communities to engage them in the conservation and, when techniques are available, in farming of seaweed species native to their coastal environments; and
e. As a reference material for various stakeholders including the Bureau of Fisheries and Aquatic Resources (BFAR), Seaweed Industry Association of the Philippines (SIAP), seaweed farmers, local government units, and others, who are involved in drafting policies on seaweed mariculture, development, and its utilization; and
f. In cases where sufficient unique and/or endemic seaweed species with enough biomass for commercial use are encountered in coastal areas close to the location of partner institutions, a special edition handbook will be prepared featuring these seaweeds highlighting their potential for food and other applications and their role in the biodiversity of the region.

2. The **standard methods that will be optimized and validated for the determination of various phytochemical properties of seaweeds** may be used in the following:
   a. As reference materials for studies in natural products and analytical chemistry by HEIs offering programs in Chemistry, Biochemistry, Biotechnology, and related programs; and
   b. As reference materials to be used by laboratories to assess quality of raw materials and processed food products derived from seaweeds and other related plant products.

3. The project will focus on phytochemical characterization of at least 50 seaweed species from the country and this may lead to identification of bioactive compounds with potential medical application. When this happens in the conduct of the PhycoPRO project, a patent application will be filed and subsequent research and development work will be performed.
   Further, when other **seaweed species that are currently not farmed are characterized to be of potential high economic value**, the **suitable cultivation techniques for this species** will be developed so that it may be utilized in mariculture activities and further processing (e.g. harvesting, biomass post-harvest processing and pretreatment, among others) of the seaweed will be studied. The cultivation techniques, including current practices and new techniques, **will be taught to new seaweed farmers to introduce them to seaweed farming to diversify their livelihood sources**.

4. The PhycoPRO project will support the establishment of two core facilities at the UP-Marine Science Institute: the ICPMS facility and Seaweed Chemistry laboratory, both of which will support not just the PhycoPRO project but also other related projects at the UP-MSI and collaborating institutions.
   In addition, the core facilities will support the following:
   a. the training of graduate students who will carry out their research and will require use of the facilities. The PhycoPRO project will involve and support thesis or dissertation research of MS/PhD students to increase the number of experts in the field; and
   b. thesis/dissertation research of interested faculty members from the partner HEIs (University of Eastern Philippines in Northern Samar, Mindanao State University Naawan, Sorsogon State University, and Partido State University) will be supported by the project through use of laboratories including the core facilities. This may lead to co-authored publications in high-impact factor scientific publications and participation in national and international conferences by researchers from partner HEIs.
   c. research initiatives of faculty/researchers at the partner HEIs that require the use of the sophisticated equipment (e.g., ICPMS, GCMS, and HPLCMS). The access to the instruments may be through research collaboration or through an agreement (e.g., MOA) that may be drafted between the UP-MSI and the partner HEIs.

5. The partner HEIs will have separate budget to support procurement of equipment and materials that will kickstart research that will be led by faculty members of the respective institutions in collaboration with the proponents of the PhycoPRO project initially, but will eventually lead to research projects that will be implemented by researchers/faculty members in the collaborating HEIs. This is envisioned to contribute, if not start, to the research culture in the partner HEIs that may lead to research productivity of faculty and researchers in the partner HEIs.

The project deliverables and milestones are also detailed further in the itemized Workplan.

D. Collaboration Plan

For a successful implementation of the grant and to attain the specific objectives towards harnessing the potential of seaweeds and its associated epiphytes, **our approach is to bring together scientists from different disciplines and utilize the expertise of each project leader/principal investigator.** Dr. Roleda is a phycologist with expertise in seaweed biodiversity, seaweed natural products and algal biofuel production, and seaweed culturing and farming. Dr. Gerwick is a distinguished biochemist with expertise in natural products chemistry and biosynthesis, and biomedicine. Dr. Rodriguez is a chemist with expertise in marine biogeochemistry, algal culture, and environmental analytical chemistry. The research fellows that will be hired for this project will be graduate students with academic background in biology, chemistry, molecular biology, and microbiology. In the US, postdoctoral fellows or research staff will conduct the proposed work, unless a predoctoral student with the apppropriate background and interests can be located to work on the project.

The proposed project is multifaceted research with each component being directed by specific PLs/PIs. The seaweed collection, identification, culture, farming, and process development for food applications will be led by Dr. Roleda. Characterization and further identification of the associated epiphytes will also be carried out with Dr. Roleda's direction. The collection of seaweed and associated epiphytes for screening of potential bioactive compounds or pharmaceuticals, and toxicological evaluations, will be under the supervision of Dr. Gerwick. In addition, Dr. Gerwick and his postdoctoral associates will also supervise the training of PH-based research fellows in the collection and extraction of seaweeds for subsequent characterization and relevant assays. The optimization and validation of various methods and procedures for determination of physicochemical parameters (lipids, fatty acids, trace metal contaminant content, etc.) will be the responsibility of Dr. Rodriguez. Necessarily, part of the research work will be carried out at the MSI and part of the work will be done at Scripps.

A planning meeting will be held on the first quarter of the implementation of the project to layout sampling plans, discuss and confirm specific tasks of each research staff, carry out training of key research team members, and identify research staff who will be responsible for meeting specific research targets. Annual meetings will also be conducted to assess the progress of the project and prepare relevant reports.

A vital part of this research is to carry out systematic sampling of seaweeds from different areas in the Philippines to facilitate identification of species with the best potential for R&D. In this regard, regional higher education institutions (HEIs) including the **University of Eastern Philippines, Mindanao State University in Naawan, Sorsogon State University, and Partido State University** will be partner HEIs. As partner HEIs, representative faculty members of the institutions will contribute by helping identify seaweed sampling locations in their respective localities, participate in the sampling and preliminary preservation or drying procedures, and facilitate the conduct of training and workshops in their respective regions. The representative faculty members will also participate in the training or workshop that will be hosted in UP-MSI on how to conduct bioassays and other relevant procedures. The partner HEIs will also contribute towards the goals of the project by establishing contact with local seaweed farmers, gathering information on how local communities utilize seaweeds, and facilitating in knowledge transfer during workshops and training especially when the use of mother language is deemed more effective. In addition, faculty members, staff, and students from these institutions may also participate in the training or workshop that will be conducted in their region. As a partner HEI, faculty and students of the institutions may further benefit from the cooperation by becoming a visiting researcher to conduct academic research at the MSI laboratory funded from this project proposal.

BENEFITS TO PARTNER HEIs

The PhycoPRO project is envisioned to be the foundation of long-term research collaborations between the UP-MSI and all partner HEIs. The research on seaweeds is planned to branch out and expand to other related research and development initiatives including seaweed biodiversity and mariculture studies at the University of Eastern Philippines, Sorsogon State University, and Partido State University, and seaweed polymer chemistry research in Mindanao State University - Naawan. These partnerships are aimed towards increasing the research productivity of the HEIs and promote global competitiveness of the faculty, researchers, and degree programs in these institutions.

The outcome that the PhycoPRO proponents would like to achieve from this partnership is for the partner HEIs to be the focal SUCs in their respective provinces for seaweed research and in training of farmers and coastal community organizations. Faculty members and researchers in partner HEIs identified will be trained on current seaweed mariculture techniques, seaweed culture maintenance, and basic physicochemical monitoring protocols. For those with existing marine biology programs and seaweed/seagrass resident scientists, e.g. Sorsogon State University and Partido State University, efforts towards more complex research initiatives will be sought. This initiative will be supported by having a separate budget for each partner HEI to fund salaries of research staff and for the procurement of equipment and materials they will require to conduct research activities in collaboration with the PhycoPRO proponents and on their own as lead researchers. While the construction and facilities required for a fully-functional seaweed nursery is dependent on the research thrust of the SUC and inhouse expertise, the SUC president and CHED Commissioner-in-Charge will consider the feasibility of establishing a seaweed nursery, and apply for separate funding as a long-term impact beyond the duration of the PhycoPRO project.

Moreover, it is hoped that the financial support for equipment provided to kickstart this collaborative research project will lead to the eventual independent research initiatives of the collaborating institutions. The equipment housed in the partner institutions secured through this funding will enable researchers in the partner HEIs to continue research on seaweeds and other research directions in the future. Moreover, research initiatives of faculty/researchers at the partner HEIs that require the use of the sophisticated equipment (e.g., ICPMS, GCMS, and HPLCMS) will be supported by providing access to the instruments through research collaboration or through an agreement (e.g., MOA) that may be drafted between the UP-MSI and the partner HEIs.

All scientific outputs from the work will be attributed to collaborating parties according to the following stipulations:

(a) Authorship of scientific outputs such as publications, technical papers, and scientific presentations will be decided according to percent contributions toward the completion of the work. Mechanics on who will be listed as authors will be decided upon during the planning meeting.
(b) The Intellectual Property Rights (IPR) Policy of the UP and UC Systems will be strictly adhered to for patents or similar forms for products and processes arising from the work.

To facilitate performance of the research tasks, biological materials and extracts from biological samples need to be sent to the UCSD counterpart and this procedure will be in accordance with the Wildlife Act (Philippine Republic Act No. 9147), which provides rules for the conservation and protection of wildlife resources and their habitats. PH-based Research Staff will be responsible for the acquisition of relevant permits and licenses from pertinent government regulatory agencies.

This project is expected to lay the foundation for a long-term collaboration between the Gerwick Lab at Scripps, the Seaweed Chemistry Lab at MSI, and the partner HEIs to ensure a strong network of national and international collaborations. It is anticipated that this project will pave the way for a continual exchange of scholars between the Institutions and result in collective efforts to tackle research questions of mutual interest not just in seaweeds and their associated epiphytes but also other marine organisms.

E. Training Plan

This project is expected to contribute to the development of a critical mass of scientists, researchers, and scientifically-trained seaweed farmers.

The PH-based research fellows that will be hired for this project (total of 4 fellows) are envisioned to finish their MS degrees (at least 2 or 3) or be mid-way in their PhD dissertations (at least 1 or 2) when the project ends after two years. These fellows are expected to learn and acquire the requisite knowledge and training in seaweed ecology, algal culturing, biochemistry, biotechnological methodologies, environmental analytical chemistry, and biochemical studies. Their experience during

the duration of the project is anticipated to improve their critical thinking and knowledge in carrying out rigorous experiments to tackle important scientific questions.

To ensure that the fellows will receive adequate training, the following requirements or activities will be imposed or carried out:

(a) the fellows will be required to enroll in a graduate program offered by UPMSI or any institution offering relevant degrees;

(b) *weekly laboratory meetings* will be conducted to ensure that the progress of fellows in terms of research tasks are monitored, that the senior project staff members may provide scientific guidance whenever needed, and that hurdles or problems are addressed as soon as recognized;

(c) *attendance to relevant meetings* will be encouraged and supported so that the fellows will have the opportunity to present their work and receive feedback from other researchers working in similar fields; and

(d) *publication of results in reputable journals* authored by the fellows will be strongly encouraged so that they will experience and learn from the process as well as develop their scientific credentials.

Workshops and training for seaweed farmers will be conducted in close collaboration with partner HEIs including the University of Eastern Philippines, Partido State University, and Sorsogon State University. A total of three (3) workshop and training sessions that will be conducted together with each partner HEI will ensure that seaweed farmers will be educated on proper harvesting and preservation procedures to maximize their financial earning capacity. For each session, it is projected that at least 50 participants from the local community and at least 20 representatives from the HEI faculty, staff, and student populations will be trained. When needed, the help of community development professionals will be sought to facilitate dialogue between scientists and seaweed farmers.

At Scripps/UCSD where it is anticipated that the workforce will mainly consist of postdoctoral fellows, meeting research objectives will be partnered with educational and training objectives. A postdoctoral fellow will be expected to participate in the various relevant seminar series at Scripps/UCSD, such at the Marine Chemical Biology Forum and the once monthly Natural Products Affinity Group. The Gerwick group has two meetings per week; one is a general group meeting communicating important information for the week, reviewing laboratory procedures and policies including those having to do with safety, and a more extensive research update presentation by one of the group members. The second meeting is a quick research update wherein each lab member updates the weeks progress, identifies problems encountered, and discussion ensues to help provide solutions. Additionally, postdoctoral fellows are required to fill out NIH-style Career Development Plans; these are discussed with the postdoctoral mentor on a twice-yearly basis to ensure that development is occurring in the desired directions.

F. Institutional Research Environment

The UP Marine Science Institute (UPMSI) is one of seven academic institutes of the College of Science, University of the Philippines Diliman. It is tasked to pursue research, teaching, and extension work in marine biology, marine chemistry, physical oceanography, marine geology, and related disciplines. The UPMSI seeks to achieve the following objectives: (a) to generate basic information necessary for optimal and sustained utilization, management, and conservation of the marine environment and its resources; (b) to provide graduate-level training and extension services to develop human resource requirements in the marine sciences; and (c) to develop appropriate and environmentally-sound marine-based technologies for industrial and economic development.

The specific objectives of this project are aligned with the direction and mandate of UPMSI. Thus, logistics and infrastructure support, which includes space to house the laboratory and all equipment, necessary for the successful execution of this project will be provided by UPMSI. The equipment procured will become part of the ICPMS facility or the Seaweed Chemistry Laboratory. The ICPMS facility will house all the sophisticated equipment including ICPMS, HPLC-MS, and GC-MS, while the Seaweed Chemistry Lab will house most of the common use equipment. The core facility may be utilized for monitoring the quality of raw and processed seaweeds as a result of the proposed project, which ensures that analytical services will be available to seaweed producers for quality control and assurance of seaweed products. The institution of an ICPMS facility will pave the way for new research directions focused on trace metal analysis and speciation of metal-containing proteins and their role in aquatic ecosystems and algal biotechnology, among many scientific questions. The Seaweed Chemistry Lab will be critical for continued research into other uses and application of seaweeds and fundamental biological and ecological questions pertaining to seaweeds and associated epiphytes.

Both of these laboratories will bring complementary expertise to other existing research laboratories in UP-MSI. Moreover, it is envisioned that the core facility for sophisticated and common use equipment will service the instrumentation requirements of other projects within UPMSI and other research laboratories within and outside the UP System and other researchers in the country through a collaborative agreement either as a research agreement or a MOA between institutions.

II. Project Results

The project will be implemented following a three-pronged approach to generate knowledge necessary for future biotechnological innovations that will address some of the health challenges in the Philippines while also providing an avenue to improve socio-economic conditions in key areas in the country. The contributions arising from this project in different aspects of development are detailed as follows.

A. Handbook of phytochemical properties of seaweeds

The major output expected from this project is the publication of a handbook of seaweeds outlining their phytochemical properties and locations where they are found. To maximize the utilization of seaweed as a source of food, food additive, and nutraceuticals, there is a need for systematic and comprehensive assessment of phytochemical properties of seaweeds available in the country. The wealth of information will pave the way for utilization of seaweeds for various applications. There is also a need to understand the influence of the symbiotic epiphytes to the properties of the seaweed raw materials and processed products. On the use of locally available seaweeds for screening of bioactive compounds and drug development, this remains unexplored and may potentially contribute novel compounds that may target different health problems. **The publication of a handbook of seaweeds outlining its phytochemical properties will serve as the starting point of future efforts toward product development for various seaweed applications including food and food additives and high-value compounds such as pigments, nutraceuticals, and pharmaceuticals.**

B. Processes

The processes that will be optimized and validated during the implementation of this project comprise different areas as indicated below:

1. Physico-chemical characterization and analysis, which includes at least 11 different analytical procedures for determination of amino acid, lipid, fatty acid content, and others.
2. Seaweed extraction and subsequent characterization and bioassay of extracts for drug development.
3. Preservation techniques for seaweeds and seaweed extracts.

These processes will be optimized to enable phytochemical analyses of the different seaweed samples. After the completion of the project, these processes will constitute the protocols that will be utilized for routine monitoring of raw materials and processed seaweed samples. In addition, the procedures may also be utilized for other researches targeting similar analytes or scientific questions.

C. Scientific outputs

The scientific outputs expected from this project include at least 4 publications in high impact peer-reviewed journals. In addition, the results are also envisaged to be presented in two (2) international and two (2) national conferences and shared with the Seaweed Industry Association of the Philippines (SIAP) to aid in the association's policy development. In each of the international conferences, at least two (2) members of the research staff are expected to participate. In each of the national conferences, at least four (4) members of the research staff are expected to participate.

D. Professional development

The project will support the career development of all research staff, as well as train seaweed farmers, towards contributing to relieve the deficit in the number of scientists and farmers dedicated on improving the seaweed industry status in the country.

E. Socio-economic impacts

One of the goals of this project is to identify the farming areas where a specific seaweed species with the optimum biochemical properties grows best. The seaweed farmers in the area will be trained on proper and sustainable farming practices to improve the quality of seaweed harvests and products resulting in a better income potential.

The activities that will be undertaken to deliver these project outputs are outlined in the Work Plan Matrix.

III. Attachments

Curriculum vitae of Drs. M. Roleda, W. Gerwick, I. Rodriguez, and L. Gerwick.

Attachments are in the file "8_IHITM_Roleda_Gerwick_Attachments_PCARI2018.pdf

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